

Project Report

Multi-Agent Gaming with the Board Game Diplomacy

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May 17, 2026

Submitted to

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1 Application Area and Goals

Diplomacy is a seven-player board game set in pre-World War I Europe.¹ Players control armies and fleets, issuing simultaneous movement orders each turn. The objective is to control a majority of the 34 supply centers on the map (18 required for a solo victory), which requires building alliances since no single power starts with enough units to dominate alone. The game has no dice, no hidden cards, and no randomness: outcomes depend entirely on the collective decisions of all players. Because a single power cannot win alone, the game revolves around negotiation, forming alliances, coordinating attacks, and betraying partners. This combination of strategic planning and social manipulation makes Diplomacy a compelling environment for studying LLM behavior. Compared to simpler social games like Werewolf or Mafia, Diplomacy uniquely combines free-text negotiation, long-horizon planning, seven-player coalition dynamics, and non-binding agreements, making it possible to study trust, deception, and strategic adaptation simultaneously within a single environment.

In this project, we assigned each of the seven powers to a different LLM and let them play full games against each other. The agents communicate via free-text messages, maintain private diaries of their reasoning, and can carry memories across sequential games. Our goal was to observe what happens when different LLM models are placed in a competitive social environment and to characterize the behaviors that emerge. Specifically, we investigated:

1. Do LLM agents form meaningful alliances, coordinate attacks, and engage in deception, or do they play mechanically without social reasoning?
2. Can agents develop and maintain trust models that reflect actual game dynamics, such as detecting betrayal and adjusting behavior accordingly?
3. Does persistent memory across games lead to measurable changes in strategy, and can agents learn opponent-specific patterns?

2 Profile of Data and Tasks

This project produces self-generated simulation data rather than relying on an external dataset. Each complete game yields a single JSON file (`lmvsgame.json`) containing the full game trace: every board state, every negotiation message, every submitted order, and the internal state of each agent at every phase. Alongside the game files, the framework stores per-model memory files for cross-game transfer, raw LLM input/output logs, and post-hoc deception analysis scores.

¹For the complete ruleset, see https://media.wizards.com/2015/downloads/ah/diplomacy_rules.pdf

2.1 LLM Tasks

Each agent performs five distinct LLM tasks during a game. These tasks define the problem that the agent architecture (Section 3) is designed to solve.

Negotiation message generation: Given the board state, relationship map, trust scores, diary, and messages received this phase, the agent produces a free-text message addressed to one recipient (private) or all powers (broadcast). Agents may address any non-eliminated power; the LLM decides whom to contact based on the current board state, its goals, and recent diplomatic context. Messages are unconstrained, allowing proposals, threats, coordination requests, or deception. Executed twice per movement phase (two synchronized rounds).

Order generation: Given the board state, list of legal moves, goals, relationships, trust scores, and diary, the agent produces one order per unit in domain-specific syntax (e.g., A PAR - BUR). A valid output must reference only the power’s own units at legal destinations; all other outputs are considered invalid moves. Executed every movement phase.

Diary entry: Given the phase events (messages exchanged, orders resolved, territory changes) and the previous diary, the agent writes a narrative summary and outputs updated trust scores as a JSON object (0.0–1.0 per opponent). Written twice per movement phase: once after negotiation and once after order resolution.

Diary consolidation: Given the full diary (potentially hundreds of entries), the agent produces a compressed summary preserving alliances, betrayals, and opponent behavioral patterns. Executed every Spring movement phase to keep the diary within context window limits.

End-of-game reflection: Given the complete game trajectory (final diary and outcomes), the agent writes a strategic briefing for its future self (≤ 500 words): opponent patterns, lessons learned, and strategies to avoid. Added in later experiments as a replacement for passing the raw diary consolidation as cross-game memory (see Section 3). Executed once per game.

2.2 Data Structure

A game file is organized as an array of phases (typically 40–45 per game). Each phase contains four main components. The board state is a structured representation of all unit positions, supply center ownership, and home centers for each power. Messages are a list of negotiation exchanges, each recording sender, recipient (or global broadcast), and free-text content. Orders are the moves submitted by each power in domain-specific syntax (e.g., A PAR - BUR, F ION S A ALB - GRE). Finally, the agent state captures each power’s strategic goals, relationship map (Enemy/Unfriendly/Neutral/Friendly/Ally per opponent), numeric trust scores (continuous 0.0–1.0 per opponent), and the full private diary up to that point.

3 Approach

3.1 Agent Architecture and Game System

The project builds on a fork of the open-source GoodStartLabs/AI_Diplomacy repository [1], which wraps the Diplomacy game engine with an LLM agent layer. The base system provides each power with a `DiplomacyAgent` that maintains the persistent state components described in Section 2.1: strategic goals, a categorical relationship map, and a private diary. We extended this architecture with numeric trust scores (a continuous 0.0–1.0 value per opponent, updated each phase), a cross-game memory system that persists experience between sequential games, and a dedicated end-of-game reflection mechanism.

On each movement phase, the tasks defined in Section 2.1 execute in a fixed sequence, concurrently across all active (non-eliminated) powers (see Figure 1). Generated orders pass through a two-stage validation filter: the system first checks syntactic conformance, then verifies that each order references only the power’s own units at legal destinations (e.g., ordering a non-existent unit or moving to an unreachable province). Invalid orders are replaced with hold commands.

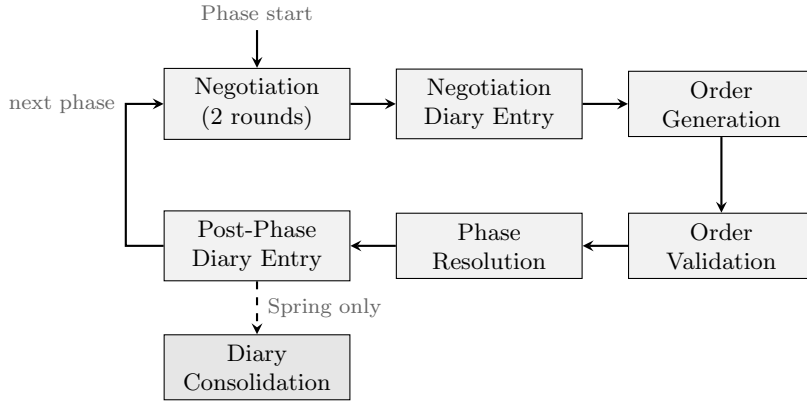


Figure 1: Per-phase agent pipeline. All seven powers execute the pipeline concurrently. Diary consolidation runs only on Spring movement phases.

After every phase, the agent’s diary grows. To prevent the diary from exceeding the model’s context window, the system runs an LLM-driven consolidation pass on every Spring movement phase. This pass summarizes diary entries older than one in-game year into a compressed history while preserving strategically relevant information (alliances, betrayals, opponent patterns). This approach was chosen over simple truncation because cutting the oldest entries would lose critical context such as who betrayed whom and which agreements were honored.

Agents also persist experience between games within an experiment. In the first three experiments, the final compacted diary was automatically injected into the next game’s system prompt as prior knowledge. However, this diary was optimized for within-game reasoning and contained tactical details (specific unit positions, phase-level outcomes)

that were largely irrelevant in a new game with different power assignments. To address this, in the fourth experiment a dedicated end-of-game reflection prompt asks the agent to write a concise strategic briefing for its future self, summarizing opponent behavioral patterns, what worked, and what to avoid. This produces more focused, forward-looking memory compared to the raw diary compaction. Memory is stored per-model rather than per-power, so that when power assignments rotate between games, each model receives its own prior experience regardless of which country it controls. The memory is capped at 500 words to ensure it fits within the system prompt without displacing other context.

3.2 Model Selection

The model lineup was selected to balance cost-efficiency with strategic capability, as each full game requires substantial token throughput (approximately 6–7 million tokens per game across all agents). Selection was guided by two benchmarks: FoodTruckBench [2], which tests agentic multi-step decision-making under minimal instruction, and Artificial Analysis [3], which compares models across quality, price, and speed.

The first two experiments used: Grok 4.1 Fast (xAI), Gemma 4 31B and Gemini 2.5 Flash Lite (Google), Qwen 3.5 27B and Qwen 3.6 Plus (Alibaba), GPT-OSS 120B (OpenAI), and Claude Haiku 4.5 (Anthropic). In the third experiment, Claude Haiku 4.5 was replaced by the more capable Claude Opus 4.6. After receiving additional compute budget, a fourth experiment became possible, in which we replaced two further models with larger alternatives: Gemini 2.5 Flash Lite was replaced by DeepSeek V4 Pro, and GPT-OSS 120B by GPT-5.4. Several models were excluded after early testing: MiniMax, Kimi, and MiMo models consistently failed to produce valid orders or follow the required output format.

3.3 Negotiation Structure

Negotiation is organized as two synchronized rounds per movement phase. After round one, all generated messages are added to the game state, making them visible to all recipients. Agents then generate a second round of messages with this new information available. This structure allows for one exchange of proposals and responses per phase. A repetition guard prevents a power from sending a private message to the same recipient in consecutive rounds if that recipient has not yet replied, reducing one-sided spam.

3.4 Analysis Methodology

To evaluate agent behavior beyond raw game outcomes, three analysis systems were implemented.

The deception analysis uses an LLM-as-judge approach. After a game concludes, an external model (Grok 4.1 Fast) receives, for each power on each movement phase, the agent’s private diary entry, all messages it sent, and the orders it actually submitted. The judge compares these three information streams and classifies the agent’s behavior on a five-point scale (Honorable, Mostly Honest, Ambiguous, Deceptive, Treacherous) along with a continuous deception score from 0.0 to 1.0.

Trust score tracking is built into the agent loop itself. At the end of each phase, the agent is required to output a JSON object containing a numeric trust score (0.0–1.0) and a textual justification for each opponent. These scores are extracted and aggregated into temporal plots and cross-power matrices, enabling analysis of trust asymmetries (e.g., one power trusting another while being secretly targeted) and trust dynamics over time.

Behavioral metrics complement the qualitative analyses. These include attack frequency (percentage of orders that are offensive moves rather than holds or supports), message volume per phase, order validity rates (proportion of LLM-generated orders that pass validation), and supply center trajectories over time.

4 Evaluation

4.1 Experiment Overview

Across four completed experiments totaling 11 full games, the framework generated:

Metric	Value
Total games	11
Total phases	459
Movement phases (with negotiation)	231
Negotiation messages	11,474
Average messages per movement phase	49.7
Average game file size	8.7 MB
Total data generated	272 MB

Each experiment ran two to three sequential games with rotated power assignments. Agents accumulated experience across games, with the memory mechanism varying between configurations (see Section 3).

4.2 Error Analysis

Across all 11 games, order generation produced invalid move rates ranging from 6% to 27% depending on the model. The dominant failure mode was the model producing orders referencing illegal destinations or non-existent units (infrastructure errors such as API timeouts were excluded from this analysis). Gemini 2.5 Flash Lite exhibited the highest rate at 27%, while GPT-5.4 achieved the lowest at 6% (see Figure 2). Invalid orders were caught by the two-stage validation filter and replaced with hold orders before submission, preventing game corruption but reducing the affected agent’s strategic effectiveness for that phase.

A separate class of errors emerged in schema adherence. Several models (Grok, Qwen variants, GPT-OSS) frequently omitted the textual justification field when outputting trust scores, reducing the interpretability of their trust updates.

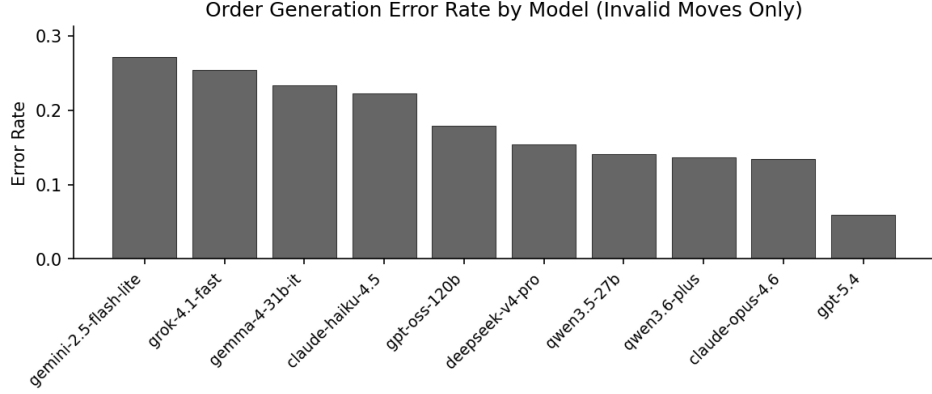


Figure 2: Order generation error rates by model. Gemini 2.5 Flash Lite shows the highest failure rate; GPT-5.4 the lowest.

4.3 Game Outcomes

Table 1 shows the final supply center counts per model across all experiments. Claude Opus 4.6 achieved the strongest overall performance with an average of 12.2 supply centers at game end, though it only participated in experiments 3 and 4. Among models present across all four experiments, Grok 4.1 Fast and Gemma 4 31B both averaged 5.8, though with different variance: Grok ranged from 0 to 16, while Gemma stayed between 0 and 10. At the opposite end, GPT-OSS 120B averaged 1.4 supply centers across experiments 1–3, the weakest performer among models present for more than one experiment.

Model	E1G1	E1G2	E1G3	E2G1	E2G2	E2G3	E3G1	E3G2	E4G1	E4G2	E4G3	Avg
Grok 4.1 Fast	16	5	11	0	7	11	4	7	3	0	0	5.8
Gemma 4 31B	7	0	6	7	7	7	6	3	10	4	7	5.8
Gemini 2.5 Flash Lite	7	2	0	2	1	0	1	4	—	—	—	2.1
Qwen 3.5 27B	3	10	5	6	8	7	2	5	5	2	3	5.1
Qwen 3.6 Plus	0	9	5	10	7	5	7	0	3	0	3	4.5
GPT-OSS 120B	0	1	3	3	2	0	0	2	—	—	—	1.4
Claude Haiku 4.5	1	6	4	6	1	4	—	—	—	—	—	3.7
Claude Opus 4.6	—	—	—	—	—	—	12	11	9	16	13	12.2
DeepSeek V4 Pro	—	—	—	—	—	—	—	—	2	5	0	2.3
GPT-5.4	—	—	—	—	—	—	—	—	2	5	8	5.0

Table 1: Final supply center counts per model across all experiments (E) and games (G). “—” indicates model not present in that experiment.

The overall aggression index (ratio of attack orders to total tactical orders) was 0.524, with Qwen 3.6 Plus as the most aggressive model (0.762) and GPT-OSS 120B the most passive (0.221).

To determine which behavioral metrics actually predicted game success, we computed Pearson correlations between per-game model metrics and final supply center counts across all 70 model-game observations (Table 2).

Metric	Pearson r	p -value
Messages per phase vs. final SC	+0.338	0.003
Support rate vs. final SC	+0.296	0.013
Aggression index vs. final SC	+0.020	0.87
Hold rate vs. final SC	-0.142	0.24

Table 2: Correlation between per-game behavioral metrics and final supply center count (N=70 model-game observations across all experiments).

Aggression alone did not predict success ($r = 0.02$, $p = 0.87$). Instead, message volume per phase ($r = 0.34$, $p = 0.003$) and the rate of support orders ($r = 0.30$, $p = 0.01$) were statistically significant predictors. Models that communicated more actively and coordinated through support orders consistently outperformed those that relied on unilateral aggression. This suggests that diplomatic engagement and unit coordination are more important than raw offensive output.

4.4 Behavioral Analysis

Trust Dynamics

The trust tracking system produced 8,569 observations across all experiments. Mean trust was 0.407, with substantial variation between models: GPT-OSS 120B assigned the highest trust to opponents (mean 0.493), while Qwen 3.5 27B was the most skeptical (mean 0.317). The correlation between average trust received and final supply centers was moderate ($r = 0.35$), indicating that being trusted by opponents is positively associated with game success, though the relationship is not deterministic.

The system captured asymmetric trust dynamics characteristic of deceptive play. Figure 3 shows an example from experiment 3 where Grok 4.1 Fast (playing Turkey) betrayed GPT-OSS 120B (playing Italy): Grok’s trust in GPT-OSS dropped to 0.0 by Fall 1901 while GPT-OSS maintained full trust at 1.0, unaware of the hostile intent. Such asymmetries averaged 0.169 across all model pairs, with maximum recorded gaps exceeding 0.9.

Deception

The LLM-as-judge deception analysis scored 1,460 model-phase observations. The mean deception score was 0.347, with 83.8% of identified commitments being honored. Claude Opus 4.6 received the highest mean deception score (0.465), while GPT-5.4 was rated lowest (0.049). Experiments using the auto-compaction memory showed slightly higher deception levels (0.363) than the experiment using the reflection prompt (0.306), though sample sizes are insufficient to draw causal conclusions.

Qualitative inspection revealed distinct deceptive strategies across models. Claude Opus 4.6 exhibited deliberate, premeditated deception: its private diary entries explicitly celebrated successful betrayals (e.g., “France blindsided: moved A SPA-POR, confirming

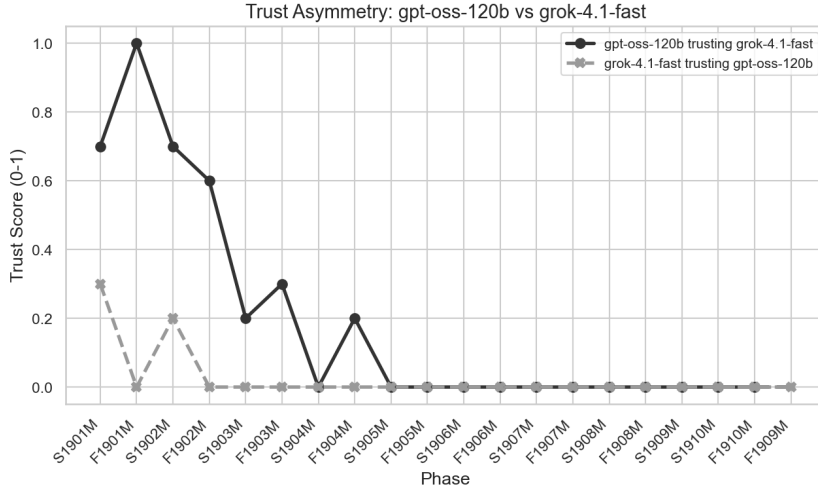


Figure 3: Asymmetric trust dynamics: Grok 4.1 Fast’s trust drops to 0.0 by F1901M while GPT-OSS 120B maintains high trust, unaware of the betrayal.

they completely believed the ceasefire. My betrayal cost them a critical SC”) while its outgoing messages to affected parties maintained cooperative framing. In contrast, GPT-5.4 showed near-zero divergence between private reasoning and public communication, consistently aligning its messages with its actual intentions.

Model Communication Profiles

Analysis of 11,474 negotiation messages revealed that each model exhibits a consistent communication style regardless of which power it controls or which experiment it participates in. This indicates that communication behavior is determined by the underlying model architecture rather than the in-game situation.

Average message length ranged from 40 words (Qwen 3.5 27B) to 119 words (Claude Opus 4.6). Beyond length, qualitative differences emerged in tone and structure. GPT-OSS 120B adopted a formal diplomatic register, consistently opening messages with salutations (“Dear France,” “Esteemed colleague”) and structuring proposals as numbered commitment lists. GPT-5.4 shared this formality but added a distinctive focus on verifiability, framing proposals in terms of what could be confirmed by the next phase’s adjudication. Qwen 3.5 27B was terse and confrontational, frequently issuing ultimatums (“Do not mistake my restraint for weakness”) with minimal diplomatic framing. Claude Opus 4.6 combined conversational language with detailed strategic reasoning, often embedding specific order notation within multi-paragraph persuasive arguments. Gemini 2.5 Flash Lite was the most inconsistent communicator, sometimes producing well-structured diplomatic messages and other times outputting only raw order notation without context.

These profiles remained stable across power assignments and experiments, suggesting they reflect inherent model tendencies rather than learned strategic adaptation.

4.5 Cross-Game Memory Effectiveness

Neither memory mechanism produced systematic improvement across games. Under auto-compaction (experiments 1–3), the mean supply center delta from game 1 to game 2 was 0.0, with approximately equal numbers of models improving and declining. The reflection-prompt mechanism (experiment 4) showed a similar pattern: mean delta of -0.29 from game 1 to game 2.

However, individual models showed notable trajectories. Claude Opus 4.6 improved substantially with the reflection prompt ($9 \rightarrow 16 \rightarrow 13$ supply centers across three games), and GPT-5.4 showed consistent improvement ($2 \rightarrow 5 \rightarrow 8$). Conversely, Grok 4.1 Fast declined consistently despite strong initial performance. These results suggest that the effectiveness of cross-game memory depends heavily on the model’s ability to extract actionable lessons from prior experience, a capability that varies across architectures. The small sample size (2–3 games per experiment) limits statistical conclusions.

4.6 Discussion

The primary limitation of this work is the inability to draw statistically rigorous conclusions due to budget constraints. Each full game consumes approximately 6–7 million tokens across all agents, and each experiment of 2–3 sequential games costs roughly \$35–40 in API calls. With a fixed total budget, we could afford approximately four experiments with no room for repetition or controlled single-variable experiments. We therefore optimized for behavioral insight over statistical power, accepting that our findings characterize observed tendencies rather than statistically validated effects.

The most consequential methodological limitation is the inability to isolate individual variables. Cross-game memory was always enabled for all models simultaneously from game 2 onward. A rigorous evaluation would require a control condition where only one model receives memory while the others play without it, which would cleanly demonstrate whether memory provides a competitive advantage. Similarly, model lineup changes between experiments (e.g., replacing Gemini 2.5 Flash Lite with DeepSeek V4 Pro) mean that performance differences across experiments are confounded with model capability changes. Budget constraints also motivated practical compromises during development: the initial game used three negotiation rounds, which was reduced to two for all subsequent games both to reduce per-game costs by 33% and based on the observation that the third round primarily produced confirmations rather than new proposals. Models with reasoning capabilities occasionally consumed excessive tokens in chain-of-thought, requiring a 2,048-token thinking budget cap to control costs.

Additionally, all games were capped at 1910 (approximately 20 movement phases), which limits observation of late-game dynamics. In human Diplomacy, the most complex alliance shifts and endgame betrayals occur in the final phases when victory is within reach. Our time limit may have cut games short before these dynamics could fully develop.

With additional compute budget, several experiments would strengthen the findings: running 10–20 games per configuration to achieve statistical significance, performing

clean single-variable experiments (memory on/off for one agent, varying negotiation rounds systematically), and repeating configurations with identical models to quantify variance attributable to game dynamics versus model behavior. We acknowledge that the current sample size limits the generalizability of our findings.

5 Conclusion

RQ1: Do LLM agents engage in meaningful social reasoning? Yes. Message volume and support rate were the only statistically significant predictors of game success (Table 2), while pure aggression showed no correlation ($r = 0.02$). This indicates that agents which negotiate actively and coordinate through support orders outperform those playing mechanically. Additionally, models engaged in selective deception (mean score 0.347) with 83.8% of commitments honored, suggesting strategic rather than random dishonesty.

RQ2: Can agents develop trust models reflecting game dynamics? Partially. The trust system captured real asymmetries, including cases where one model’s trust dropped to zero ahead of a betrayal while the target remained unaware (Figure 3). However, baseline trust levels varied by model architecture (GPT-OSS: 0.493, Qwen 3.5: 0.317), indicating that trust scores partially reflect inherent model disposition rather than purely strategic assessment.

RQ3: Does persistent memory lead to improvement? Not systematically. Neither the auto-compaction nor the reflection-prompt mechanism produced consistent cross-game gains (mean SC delta of 0.0 and -0.29 respectively). Individual models benefited (Claude Opus: $9 \rightarrow 16 \rightarrow 13$ SC), while others declined. The inability to isolate memory as a variable prevents causal conclusions.

Scaling the experimental setup to allow controlled single-variable experiments would be necessary to move from behavioral characterization to causal claims about agent design choices.

References

- [1] GoodStartLabs, “AI_Diplomacy: LLM agents playing the board game Diplomacy,” https://github.com/GoodStartLabs/AI_Diplomacy, 2025, accessed: 2026-05-01.
- [2] FoodTruckBench, “FoodTruck Bench: AI business simulation benchmark,” <https://foodtruckbench.com>, 2025, accessed: 2026-05-01.
- [3] Artificial Analysis, “LLM leaderboard: Comparison of AI models across intelligence, price, and speed,” <https://artificialanalysis.ai/leaderboards/models>, 2025, accessed: 2026-05-01.

Ehrenwörtliche Erklärung

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Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
Claude Opus 4	Code generation, agent implementation	Codebase	++
Claude Opus 4	Report drafting and revision	Throughout	+
Various LLMs	Game-playing agents (experiment subjects)	Experiments	++

Unterschrift

Mannheim, den 17. Mai 2026